

Employees' Behavioral Intention to Adopt Facial Recognition Payment to Service Customers: From SQB and Value-Based Adoption Perspectives

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ABSTRACT

Facial recognition payment (FRP) is a crucial technology for digital transformation among merchants. However, few studies have explored employees' adoption of FRP. This study combined the status quo bias theory and the value-based adoption model to investigate the critical factors affecting employees' behavioral intention of FRP, including switching benefits, job insecurity, colleague opinion, top management support, and inertia. The results show that switching benefits, job insecurity, colleague opinion, top management support, and inertia are essential factors that determine employees' behavioral intention of FRP. Surprisingly, the perceived customer value negatively moderates the relationship between perceived value and behavioral intention. Furthermore, the authors provide rigorous endogeneity tests for the research model. Overall, this study contributes to the status quo bias and value-based adoption model theories and literature in organizations' digital transformation and offers valuable insights to eliminate resistance and encourage the active use of FRP.

KEYWORDS

Behavioral Intention, Employee, Facial Recognition Payment, Status Quo Bias, Value-Based Adoption Model

INTRODUCTION

In 2019, facial recognition payment (FRP) emerged as a novel payment method for the commercial market. Since then, FRP has been rapidly expanding its presence in the market. After four years of development, FRP has garnered a substantial user base, numbering in the hundreds of millions (iiMedia Research, 2019). Concurrently, FRP systems have seen widespread adoption by numerous offline industry entities, including those operating in retail, catering, medical and health, and education sectors. For these organizations, using FRP systems offers multifaceted benefits, encompassing reduced procurement expenses, heightened payment efficiency, diminished customer queue times, and

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sporadic red-envelope incentives extended to customers by system providers (e.g., Alipay, WeChat Pay, UnionPay). As a terminal carrier, FRP could be a pivotal innovation for merchants' digital upgrades (Mu, 2021). For instance, FRP systems could provide merchants with a powerful business platform as a service product. The business component platform of FRP offers a membership system without the need for development and facilitates easy access to various value-added services for customers. Additionally, FRP devices could enhance digital store operations, improving essential functions such as store flow, business analysis, and prediction. The ability to capture detailed customer profiles facilitates organizations in delivering tailored services and enhancing levels of customer relationship management (Yinong Network, 2020). Moreover, FRP systems are expected to evolve into intelligent IoT business platforms (Fang & Zhang, 2024). As an entry point to merchant digital services, FRP can help better integrate online and offline interactions. In the future, it is poised to become an indispensable component of the transition to an intelligent world. Consequently, integrating FRP systems within physical enterprises represents a shift in payment methodologies and an information technology avenue facilitating digital business transformation (Shiau et al., 2023; Zhong & Moon, 2022).

Digital transformation constitutes a strategic imperative, necessitating all employees' comprehensive adoption and implementation of specific information systems (IS), particularly frontline staff. The efficacy of an organization's digital transformation endeavor and the seamless execution thereof hinges fundamentally on employees' willingness to embrace and operationalize novel technologies and systems (Huang et al., 2023; Shahreki et al., 2024; Timbula & Marvadi, 2023). Prior scholarly inquiries have delved into employees' proclivity for either resisting or supporting new technologies, with resistance to innovation attributed to nearly 70% of organizational setbacks in implementing innovative initiatives (Shahbaz et al., 2019). Factors underpinning employee resistance to novel technologies may stem from perceived job insecurities (Feng et al., 2021), inherent aversion to adapt to new technological paradigms (Laumer et al., 2016), or reliance on entrenched technological modalities (Wang, 2022). According to Frey and Osborne (2017), approximately 47% of jobs in the United States are anticipated to undergo automation within 10–20 years. Conversely, alternative research posits that employees exhibit enthusiastic responses and express satisfaction upon witnessing technology's replacement of manual and repetitive tasks (Lacity & Willcocks, 2016). These disparate findings underscore the intricate cognitive dynamics characterizing employees' attitudes toward technological advancements. For instance, the function and value of FRP systems introduced by the organization depend on the attitude of employees toward this system. In other words, whether it is successfully used after introduction, rather than becoming a technology system lacking practicality, hinges on whether employees strongly intend to switch from prioritizing other payment methods to using FRP systems to serve customers. Employees may have mixed perceptions about the introduction of FRP systems. Due to status quo bias (SQB), employees tend to remain resistant to change (Chatterjee et al., 2022). Additionally, manual replacements, layoffs, and transfers caused by FRP systems may further negatively affect employees. Meanwhile, digital human-computer interaction in FRP systems will improve employee efficiency and simplify operational processes, enhancing employees' sense of value. The individual adoption of FRP has garnered significant scholarly attention, with numerous studies investigating various facets of this phenomenon. For instance, some research has explored the antecedents and mechanisms influencing individual consumers' intentions to adopt FRP (Ciftci et al., 2021; Gao et al., 2024; Hwang et al., 2024; Kim et al., 2024; Lee & Pan, 2022; Li & Li, 2023; Lim et al., 2024; Shiau et al., 2023). Conversely, other studies have examined privacy-related concerns among individual consumers who resist FRP adoption, delving into the more challenging aspects of this issue (Lee & Pan, 2023; Zhang & Zhang, 2024). The collective findings of these studies have significantly advanced our understanding of the formation mechanisms and influencing factors underlying individual consumers' intentions to use FRP. However, scant attention has been devoted to examining the attitudes and intentions of employees toward FRP.

To address this gap in the literature, the present study seeks to identify the primary factors that either foster or hinder the behavioral intentions of employees in offline organizations utilizing FRP systems, focusing on the perspectives of SQB theory and perceived value. To fulfill this objective, two key research questions are posited as follows:

- RQ1. What are the critical factors influencing employees' behavioral intention to adopt FRP systems in offline organizations as payment technology to provide customers with services?
- RQ2. How does perceived customer value addition moderate the relationship between perceived value and behavioral intention on adopting FRP systems?

The study's contributions lie in three aspects. Firstly, it integrates the SQB theory and value-based adoption model (VAM) theoretical framework to advance the literature on FRP by identifying the crucial factors that affect employees' perceived value and behavioral intention in offline organizations. This integration provides a novel theoretical framework and perspective for researchers examining how merchants introduce alternative IS. Secondly, the study explores the moderating effect of perceived customer value addition on the relationship between perceived value and behavioral intention, explaining the formation of employees' intentions. Thirdly, the study employs a rigorous endogeneity test to identify possible endogenous variables within the research model, thereby ensuring this study's rigor.

The remainder of this paper is organized as follows: The literature review section discusses the theoretical background of FRP, SQB theory, and VAM. The research model and hypotheses section presents the processes of theorizing the model and building hypotheses. The research methodology section details the data collection and development of measurement. The results section presents the data analysis and results. The discussion section considers this study's key findings, implications, and limitations. Finally, the conclusion summarizes the study's core findings and contributions.

LITERATURE REVIEW

FRP

Facial recognition technology is powered by artificial intelligence, three-dimensional structured light biometrics, cloud service technology, and big data. It employs deep learning algorithms to collect facial feature information from images or videos, automatically detect and track faces, and authenticate individual identities (Cheng et al., 2023; Lee et al., 2013). Specifically, this technology operates through five steps: image acquisition, facial detection, feature extraction, database comparison, and identity confirmation (Cheng et al., 2022; Du et al., 2022; Laka et al., 2021; Zhang & Zhang, 2024).

Facial recognition technology has been widely utilized across various scenarios. For example, in the security field, it is used to unlock personal information on mobile phones, computers, and bank accounts, significantly enhancing information security. In public services, it accelerated the use of public transport services and improved security during inspections. In the financial sector, FRP, grounded in facial recognition technology, represents an emerging mobile payment modality that leverages human biometrics for personalized identification. The FRP device links the collected images to a designated digital payment system or bank account and compares them with those stored in the database (Abrahão et al., 2016). Characterized by a streamlined process typically spanning 10–15 seconds and predominantly interfacing with automated systems, FRP stands poised for widescale commercial implementation within offline enterprises, propelled by the ongoing evolution of digital technologies. Alipay and WeChat Pay collaborate with different suppliers at the application layer, resulting in hundreds of payment devices. Moreover, the FRP systems are integrated with other application systems, contributing to a more complex intelligent service industry. This includes

intelligent retail cabinets, unmanned retail stores, smart carts, unmanned hotels, and automated plate machines.

The rapid proliferation of FRP across diverse market sectors has attracted considerable scholarly interest, with its commercial availability dating back to 2019 and research inquiries intensifying since 2021. Notably, scholarly investigations have scrutinized FRP applications from technical vantage points (Feng & Chen, 2022; Yang, 2020). The efficacy and utility of FRP systems introduced within organizational contexts hinge upon employees' reception and adoption. In essence, successful utilization post-implementation, instead of relegation to a technologically redundant system, pivots on employees' emotional inclination to transition from conventional payment methods to FRP facilitation in customer service realms. However, the likelihood of FRP adoption remains uncertain in instances where extant payment methodologies, such as quick code payments, persist alongside FRP implementation. Furthermore, extant literature predominantly addresses consumer-centric facets, delving into determinants of individual consumers' propensity for adoption or resistance (Gao et al., 2024; Liu & Tu, 2021; Shiau et al., 2023), with scant attention devoted to frontline employee utilization within organizational settings.

Diverging from consumers' experiential considerations, employees' engagement with FRP is contingent upon individual psychological factors and vocational performance metrics (Chatterjee et al., 2022; Shiau et al., 2023). Unlike consumers who utilize FRP for shopping and leisure, employees deploy FRP to cater to clientele and facilitate workplace transactions, thereby engendering distinct evaluative criteria. During FRP promotion's nascent stages and transition from contact-based to contactless payment paradigms, employees exhibit a keen interest in usage intent and efficacy as conduits between organizations and consumers and direct operators of FRP systems. Moreover, select scholars have invoked the SQB theory to elucidate employees' technical resistance to digital transformation (Chatterjee et al., 2022; Li et al., 2016; Oschinsky et al., 2021), while the VAM theoretical framework has been deployed in mobile payment research to elucidate the nexus between individual values and intent (Jun et al., 2018). Notably, these theories are seldom integrated to explore the adoption of the FRP system. Consequently, this study integrates SQB and VAM theories to scrutinize employees' behavioral inclinations toward utilizing FRP to serve customers.

SQB Theory

The SQB theory elucidates why individuals persist with an incumbent course of action rather than transitioning to a new (potentially superior) course (Samuelson & Zeckhauser, 1988). It furnishes valuable theoretical insights for comprehending the impact of incumbent system use as a deterrent to embrace new systems (Polites & Karahanna, 2012). Samuelson and Zeckhauser (1988) delineate SQB explanations as encompassing rational decision-making, cognitive misperceptions, and psychological commitment. The dimension of rational decision-making under uncertainty requires the individual to assign probabilities to possible outcomes and calibrate utilities to evaluate those outcomes (Savage, 1954). It implies assessing a change's relative costs and benefits before switching to a new alternative. Cognitive misperceptions can arise when individuals weigh potential losses from switching as greater than potential gains, leading to biased decision-making due to loss aversion. Consequently, individuals tend to favor the status quo. SQB can also be explained by psychological commitment. Individuals often seek to justify previous commitments by making additional follow-up commitments. The greater the past resource investment in a decision, the stronger the tendency to continue committing to subsequent decisions (Samuelson & Zeckhauser, 1988).

Studies on SQB theory have garnered increasing attention in the field of IS (Gong et al., 2020; Wang et al., 2018). This theory posits the rational decision-making process of users in adopting or rejecting an information technology system (Balakrishnan et al., 2021; Talukder et al., 2021). Kim and Kankanhalli (2009) applied SQB to investigate individual resistance to innovation. In individual-focused research, most scholars study technology acceptance among individuals (Chakraborty et al., 2024; Lee and Li, 2023). A minority of scholars use SQB theory to analyze the

willingness or resistance of employees to innovative technology as individual users in the enterprise. For example, Alzahrani et al. (2021) examined employees' resistance and mental readiness to accept enterprise resource planning (ERP) systems in the pre-adoption phase. Chatterjee et al. (2022) explored the influence of individuals' skills and capabilities on the success of digital transformation in organizations utilizing AI customer relationship management. Li et al. (2016) investigated why employees resist knowledge management systems. In the context of mobile payment, SQB theory is utilized to examine various aspects, including innovation resistance in mobile payment adoption and usage (Kaur et al., 2020), the impact of new technologies on the success of digital payment (Ferguson & Neubert, 2019), the reasons behind the slow adoption of mobile payment from a switching intention perspective (Loh et al., 2021), the effect of inertia on users' intention to use new mobile payment services (Gong et al., 2020), and the formation of switching intentions for mobile payment services platforms (Kuo, 2020). The prevailing body of literature addresses individual resistance toward adopting specific mobile payment technologies, the factors influencing individuals' propensity to adopt mobile payment methods, and their motivations for transitioning from cash-based transactions to mobile payment alternatives.

SQB can often be combined with other models to broaden the application scope and enhance the explanatory power of these frameworks. For instance, it can be integrated with the push-pull-mooring model (Kuo, 2020; Loh et al., 2021; Wang et al., 2019) or the adoption model (Balakrishnan et al., 2021; Hsieh, 2015; Kim & Kankanhalli, 2009). As Samuelson and Zeckhauser (1988) demonstrated, the theoretical mechanisms underlying SQB are context-specific (Polites & Karahanna, 2012).

VAM

Kim et al. (2007) proposed the VAM by extracting and summarizing the literature concerning individuals' decision-making to elucidate the intentions and behaviors of user acceptance of IS. VAM is grounded in prospect and mental accounting theories, focusing on maximizing benefits when using new technologies (Chung & Koo, 2015). VAM explains individuals' decision-making process through a cost-benefit paradigm by comparing the costs (sacrifices) and benefits of new technologies or products (Lin et al., 2012). Sacrifices can include monetary and non-monetary aspects, while benefits encompass cognitive and emotional gains. VAM emphasizes an overall assessment of a product's utility based on perceptions of what is received versus what is given. It considers benefits and sacrifices as critical factors contributing to perceived value, defined by balancing benefits and costs and thus predicting individuals' adoption intentions. Introduced initially within the context of mobile internet and mobile commerce, VAM underscores the impact of individual users' comprehensive evaluation of the benefits and sacrifices of new technologies on their adoption intentions, particularly emphasizing decisions based on maximizing value (Kim et al., 2007). With the rapid advancement of information technology, VAM has been applied to explicate individuals' adoption of various technology systems, including AI technology (Dhiman et al., 2023; Sohn & Kwon, 2020), IoT services (Kim et al., 2017), service robots (Hong et al., 2023; Kang et al., 2023), bring-your-own-device (Lian, 2020), augmented reality technology (Yoon & Oh, 2022), medical mobile platforms (Xiong & Zuo, 2022), contact tracing mobile applications (Cocosila et al., 2022), and mobile payments (Mallat, 2007).

VAM exhibits considerable versatility and adaptability in problem-solving and interpretation, with subsequent scholars expanding the selection of variables related to benefits and sacrifices to suit specific contextual needs, thereby enriching the framework's value (Cocosila et al., 2022; Yoon & Oh, 2022).

RESEARCH MODEL AND HYPOTHESES

This study integrates VAM to explain employees' intention to adopt FRP from a value maximization perspective (Sirdeshmukh et al., 2002). VAM predicts the perceived net benefits associated with the new change. Employees who perceive a high value in the new technology or system will likely

exhibit lower resistance and higher adoption behavior (Kim et al., 2007). From a comprehensive economic and non-economic perspective, SQB theory employs economics, psychology, and decision theory to better understand employees' perceived value and adoption of FRP systems as alternative payment methods (Kim, 2011; Samuelson & Zeckhauser, 1988). SQB theory encompasses various explanatory categories, including rational decision-making, psychological commitment, and cognitive misperceptions. Rational decision-making aligns with the net benefits concept in VAM, involving evaluating essential benefits and sacrifices. In contrast, cognitive misperceptions and psychological commitment influence irrational decisions, providing further insight into the complex psychological factors affecting willingness beyond rational value analysis.

Under the rational decision-making framework, service users are expected to maximize their switching benefits. This implies that employees might prefer to use the FRP system to serve customers, even if other existing payment methods are still available (Zeithaml et al., 1988). On the other side, digital transformation may result in some level of workforce reductions (e.g., downsizing, mergers, acquisitions), which can expose employees to workplace uncertainty (Brougham & Haar, 2020; Dengler & Gundert, 2021; Mirbabaie et al., 2022). Job insecurity is defined as a sense of powerlessness to maintain desired continuity in a threatened job situation (Hellgren et al., 1999). It is a significant source of distress and anxiety for employees.

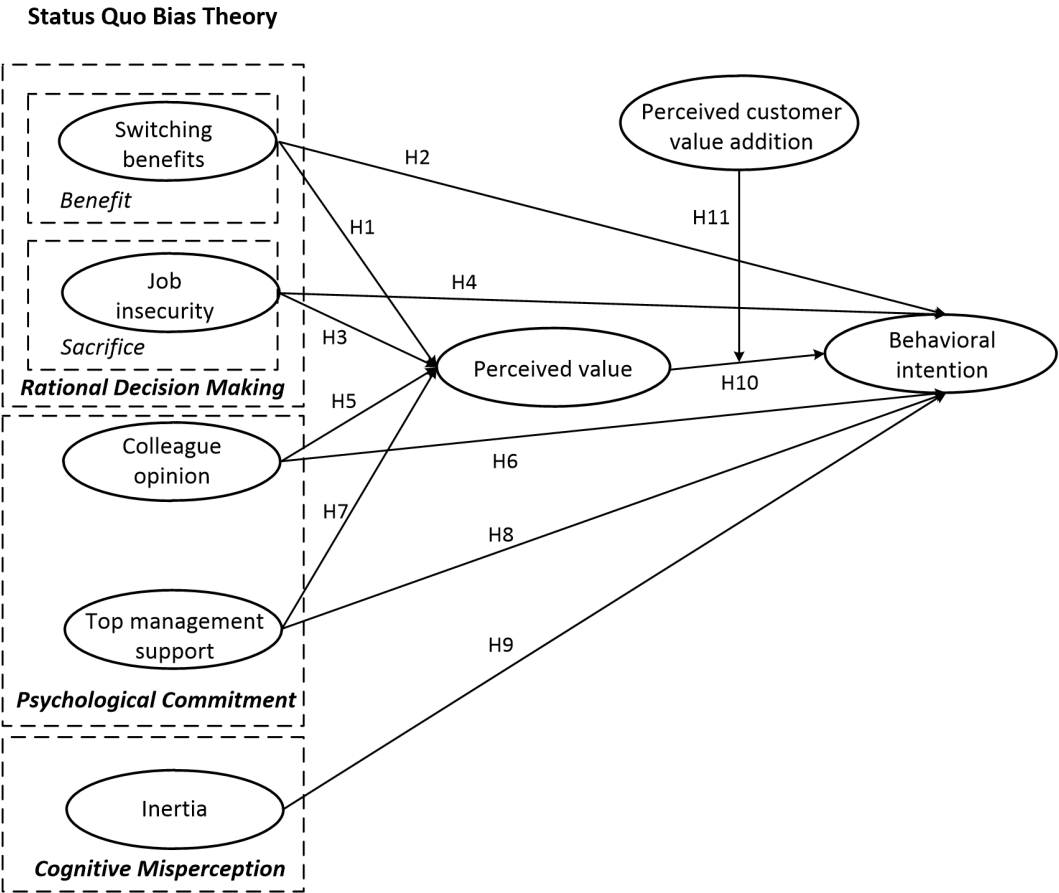
Regarding psychological commitment, social norms and efforts to feel control are two key factors related to psychological commitment (Samuelson & Zeckhauser, 1988). Colleague opinion is considered a prominent social norm of individual identity (Lewis et al., 2003) since colleagues are often important references for individuals on work-related issues (Kim & Kankanhalli, 2009). Colleague opinion is defined as the perception that colleagues favor the changes related to the FRP system (Kim & Kankanhalli, 2009), which is used to justify previous commitments to action by making follow-up commitments in the work environment. Moreover, in the digital transformation of organizations, the introduction of new equipment and systems is often driven by decisions made by top management. Therefore, the primary source of employees' sense of control within the organization stems from support from top management. Top management support is defined as the perceptions and actions of top officials on the usefulness of FRP in creating values for the organization (Intan Salwani et al., 2009).

Furthermore, Samuelson and Zeckhauser (1988) suggested that the cognitive misperception of loss aversion also explains SQB since individuals weigh losses heavier than gains in making decisions. Inertia often arises after ex-post losses if individuals are loss averse. Inertia, from the perspective of cognitive misperception, refers to a user's attachment and adherence to existing IS despite the availability of better alternatives or motivations for change (Loh et al., 2021; Polites & Karahanna, 2012). Cognitive misperceptions and psychological commitment are additional sources influencing irrational decisions. These complex psychological factors further supplement the impact of rational value analysis on the formation of willingness. Based on SQB and VAM theories, the most critical factors influencing employees' behavioral intention to adopt FRP include switching benefits, job insecurity, colleagues' opinions, top management support, and inertia. Based on the discussion mentioned above, the research model is proposed, as shown in Figure 1. The relationships and hypotheses are discussed in detail as follows.

Switching Benefits, Perceived Value, and Behavioral Intention

Switching benefits refer to the perceived utility a user would enjoy in switching from the status quo to the new IS (Kim & Kankanhalli, 2009). The switching behavior may improve an individual's performance (Mahmud et al., 2017). Then, the perceived value of this change will increase due to the potential for improved performance and the corresponding rewards. Samuelson and Zeckhauser (1988) argued that perceived value reflects the extent to which conversion benefits justify deploying new technologies. Some previous studies proposed that increasing the benefits of switching to a new cloud ERP system will increase the perceived value of using such a system (Mahmud et al., 2017).

Figure 1. The proposed research model



Similarly, Kim and Kankanhalli (2009) suggested that expanding the switching benefits of using new enterprise portals and knowledge management systems contributes to increasing the perceived value. When individuals switch to new systems, such as high job performance and efficiency, the potential benefits will increase their willingness to switch to the new system. Park and Ryoo (2013) and Chang (2020) found that switching benefits positively affect the intention to switch to cloud ERP systems.

In this study, switching benefits represent the perceived utility employees would enjoy when switching from other payment systems to FRP systems. For instance, it enhances effectiveness, increases service productivity, and improves service quality when serving customers. These potential benefits could increase the employees' perceived value and intention to use the FRP systems. Accordingly, the following hypotheses are proposed:

- H1. Switching benefits positively affect perceived value.
- H2. Switching benefits positively affect behavioral intention.

Job Insecurity, Perceived Value, and Behavioral Intention

Job insecurity is often considered the most common form of uncertainty in the workplace, as it signifies the potential loss of employees' roles due to technological advances that can automate or replace key tasks performed by humans. Employees who fear losing their jobs may experience stress due to anticipation of future issues (Greenhalgh & Rosenblatt, 1984; Vander Elst et al., 2014). The

concern about being fired is associated with nervousness, which can significantly affect an employee's mood, attitude, and behavior (Feng et al., 2021). Job insecurity is expected to be associated with negative reactions from employees, such as decreasing job satisfaction levels (Aslam et al., 2016) and changing affective commitment (Feng & Chen, 2022). In the context of technology-related workplace change, Mirbabaie et al. (2022) examined how employees' job insecurity leads to identity threats when AI is introduced into the workplace. Lee et al. (2022) suggested that organizational change caused by technological advancements may induce job insecurity and then result in negative work outcomes. In the case of FRP systems, since FRP systems are more cost effective than other mobile payments, many retail stores and other organizations prefer FRP systems to replace employees. The remaining employees may experience instability, ambiguity, and unpredictability, which can diminish their perceived value of FRP systems (Kim & Kankanhalli, 2009). Furthermore, job insecurity often leads to lower work engagement, which reduces the intention to use FRP systems. Thus, the following hypotheses are proposed:

H3. Job insecurity negatively affects perceived value.

H4. Job insecurity negatively affects behavioral intention.

Colleague Opinion, Perceived Value, and Behavioral Intention

Colleague opinion is a significant social influence that an individual identifies within the work environment (Lewis et al., 2003). In the workplace, individuals tend to agree positively or negatively with their colleagues' perceptions of new information technologies due to the need for social companionship (Lewis et al., 2003). Thus, colleagues are suggested as significant references for influencing personal opinion, and individuals may follow their colleagues to decide on their attitude toward the behavior of use (Kim & Kankanhalli, 2009; Seol et al., 2016). For instance, Goyal et al. (2022) suggested that the perceived value of an online doctor consultation platform will largely depend on the social impact created by peers. Huang (2015) found that colleagues' opinions increase the perceived usefulness of employees learning using advanced computer software. Additionally, subjective norms in the workplace represent an employee's opinion on whether a vital reference person approves or disapproves of a specific behavior (Ajzen, 1991). Kim and Kankanhalli (2009) found that beneficial colleague opinion may reduce user resistance to new technologies, i.e., colleague opinion positively affects users' willingness to adopt new IS. Similarly, Seol et al. (2016) found that colleague opinion is closely and positively related to employees' intention to use mobile offices. In FRP systems, individuals will evaluate the value of new payment instruments based on colleagues' opinions. The perceived value will increase if colleagues' feedback is a favorable opinion. Moreover, group members often feel pressured to conform to group behavior patterns. Thus, colleague opinion is considered an essential reference in the work scene, and it will affect the behavioral intention of individual employees to prioritize using the FRP system. Therefore, the following hypotheses are presented:

H5. Favorable colleague opinion positively affects perceived value.

H6. Favorable colleague opinion positively affects behavioral intention.

Top Management Support, Perceived Value, and Behavioral Intention

Top management support refers to the motivation and encouragement provided by management for the adoption of IS (Ifinedo, 2011). Top managers often play a crucial role in influencing and persuading other members to embrace new technologies (Kumar et al., 2017). Support from top management significantly affects an organization's perception of the actual use of these technologies (Lin, 2010; Pizam et al., 2022). Support from top management will increase individuals' perception of the benefits of technology. Thatcher et al. (2006) suggested that support from top management will encourage individuals to understand the benefits of adopting a new approach. As a result, the odds of successful adoption of IT/IS will increase dramatically. In addition, top management support may also give employees encouragement and support (Hwang et al., 2021), dispel concerns and psychological

burdens through good communication (Maduku et al., 2016), and increase employee training (Wang & Lai, 2014), thereby generally improving employees' awareness of the value of technology products. Ifinedo (2008) argued that when top management support for an IT/IS implementation project is widely known among organizational members, employees are more likely to perceive IT/IS implementation as a positive initiative and, thus, are more likely to engage in the process. Moreover, the support of top management reflects the top-down organizational environment atmosphere and robust control. Although the acceptance of technology is not mandatory, the attitude and sense of support from their superiors will also have a significant impact on employees' willingness to adopt technology (Wang & Lai, 2014). The decision of the organization to introduce the FRP system usually comes from the top. This decision may involve the adjustment of employee positions, the number of employees, redesigning the service process, and job content. Therefore, the specific support methods of top management will enhance employees' positive awareness of benefits and sacrifices, improve employees' perception of value for FRP systems, and reduce acceptance barriers, thus making them easier to adopt. Meanwhile, the leadership demonstrated by management's strong support and excellent attitude can also directly lead to greater acceptance of the FRP system. Thus, the following hypotheses are presented:

H7. Top management support positively affects perceived value.

H8. Top management support positively affects behavioral intention.

Inertia and Behavioral Intention

The results of individual inertia are like those induced by structural state dependence (Dube et al., 2009) and attitude loyalty (Lee & Neale, 2012), except that individual inertia achieves its effects in another way. Individuals tend to maintain the status quo by developing inertia due to cognitive errors in new technologies (Samuelson & Zeckhauser, 1988). Since individuals have adapted to the current system, they will feel comfortable working with the existing system and will be very skeptical about adopting new technology (Recker, 2014). Some studies argued that inertia is a crucial indicator of an individual's resistance behavior and intention (De Mesquita & Urdan, 2019; Polites & Karahanna, 2012; Shankar & Nigam, 2021). In addition, uncertainty about new technologies plays a crucial role in generating personal inertia, leading to resistance to adopting new technologies (Seth et al., 2020). After the organization introduces an FRP system as a new payment method, employees need to decide whether to abandon the old payment system that they have used smoothly, highly accepted, and been satisfied with and use the new system as a priority payment tool, which also includes recommending and guiding customers to use it. In addition to changing usage habits, employees also need to face the potential loss caused by the uncertainty of using the new system and communicating and interacting with customers in customer service. Therefore, the inertia shown by employees to rely on the original payment method will make them less willing to use FRP systems. Accordingly, the following hypothesis is proposed:

H9. Inertia negatively affects behavioral intention.

Perceived Value and Behavioral Intention

Zeithaml et al. (1988) argued that perceived value is an individual's overall evaluation of what is received and given. It reflects an individual's subjective, individualized evaluation of product quality or service (Hapsari et al., 2016). It can represent the judgment of choosing a strategy and influence an individual's decision-making behavior. As a crucial indicator of intention, perceived value has been investigated in the context of IS (Xie et al., 2021). In mobile payment, researchers have found that individuals' willingness to implement new systems is highly correlated with perceived value (Sun et al., 2022). In this study, perceived value refers to the employee's evaluation of the FRP system as a payment method to serve customers based on the perception of sacrifice and benefit trade-off. Employees often act as recommenders and mentors of checkout methods and have the preferred choice of payment methods. The importance of providing services based on values is emphasized

for the option of using the FRP system, so the perceived value will be an essential factor for them to consider when using this new payment system. Accordingly, the following hypothesis is proposed:

H10. Perceived value positively affects behavioral intention.

Perceived Customer Value Addition, Perceived Value, and Behavioral Intention

Perceived customer value addition is defined as the level of satisfaction of a service's customer, which is based on the customer's emotional, physical, and social response and perceived rating given to the service or retailer (Singh & Sinha, 2020). Brick-and-mortar organizations prefer technology valued by customers and provide comprehensive and satisfactory services (Singh & Sinha, 2020). Perceived customer value addition reflects the perspective of a third party (Singh & Sinha, 2020). Therefore, it is not an objective measure of customer value addition but rather the merchant or employee's perception of the value added to the consumer (Hayashi, 2009; Singh & Sinha, 2020). Several studies have found that employee alignment with customer attitudes increases employee motivation. For example, integrating customer satisfaction with a task or project will increase employee intention (Frey et al., 2013). Furthermore, Chen et al. (2013) found that employees who create relationship value with clients enjoy their jobs more and are more likely to build and maintain long-term relationships with institutions. That is, the consistency between the perceived value of employees and the perceived addition in customer value will increase the willingness of employees to adopt new technology. In the context of offline organizations providing services, employees are more likely to use the FRP system when they perceive that this new technology enhances their job performance and offers customers greater convenience, new experiences, and improved interaction through their recommendation or encouragement of the system. Accordingly, the following hypothesis is proposed:

H11. Perceived customer value addition positively moderates the relationship between perceived value and behavioral intention.

RESEARCH METHODOLOGY

The data in this study were examined using IBM SPSS and partial least squares (PLS) analysis (Ringle et al., 2024). SPSS (version 25.0) was used to assess the samples and help present the descriptive information of respondents. PLS structural equation modeling (PLS–SEM) was used to estimate the research model. Compared with covariance-based SEM, PLS–SEM has minimal requirements in accordance with measurement scale, sample size, and residual distributions (Chin et al., 2003) and is also suitable for complex models, with no restrictive assumptions of normal distributions and formative models (Hair et al., 2019; Shiau et al., 2019, 2020; Wold, 1982). This study's research model is complex and has many constructs, indicators, and relationships. Thus, PLS–SEM using SmartPLS version 4.1 (Ringle et al., 2024) is appropriate for this study.

Measurement Development

This study proposes a model for investigating employees' intention to use FRP systems to serve customers in offline organizations. The operational definitions of all variables and scale items were adapted from the mature scale. Certain adjustments and modifications were made according to the research context while preserving the original measurement focus of the scales. The items of eight constructs were scored on a 7-point Likert scale (ranging from strongly disagree = 1 to strongly agree = 7). Table 1 presents all operational definitions of the constructs and scale items adapted from the related literature.

The questionnaire focuses on the factors and mechanisms that affect employees' intention to use FRP. “A Survey on Offline Employees' Intention to Provide FRP Services to Consumers” is divided

Table 1. Operational definitions of constructs and measurement items

Constructs	Measurement Items	Adapted from
<i>Switching benefits</i> Switching benefits refer to the perceived utility a user would enjoy in switching from the status quo to the new IS (Kim & Kankanhalli, 2009).	SWB1 Changing to the new way of working with the FRP system would enhance my effectiveness on the job than working in the current way.	Kim & Kankanhalli (2009)
	SWB2 Changing to the new way of working with the FRP system would enable me to accomplish relevant tasks more quickly than working in the current way.	
	SWB3 Changing to the new way of working with the FRP system would increase my service productivity than working in the current way.	
	SWB4 Changing to the new way of working with the FRP system would improve the quality of the work compared to the current method.	
<i>Job insecurity</i> Job insecurity refers to a sense of powerlessness to maintain desired continuity in a threatened job situation (Hellgren et al., 1999).	JIS1 If the FRP system is introduced, I am worried about having to leave my job before I would like to.	Mirbabaie et al. (2022)
	JIS2 If the FRP system is introduced, there is a risk that I will have to leave my present job in the year to come.	
	JIS3 If the FRP system is introduced, I feel uneasy about losing my job soon.	
	JIS4 If the FRP system is introduced, my future career opportunities in the organization are not favorable.	
	JIS5 If the FRP system is introduced, I believe that the organization will not need my competence also in the future.	
	JIS6 If the FRP system is introduced, my salary development in this organization is not promising.	
<i>Colleague opinion</i> Colleague opinion is defined as the perception that colleagues favor the changes related to the FRP system (Kim & Kankanhalli, 2009).	CO1 Most of my colleagues think the change to the new way of working with the FRP system is a good idea.	Belanche et al. (2022); Kim & Kankanhalli (2009)
	CO2 My peers are supportive of the change to the new way of working with the FRP system.	
	CO3 Most people whom I deal with in my job encourage me to change to the new way of working with the FRP system.	
	CO4 My peers, who are important to me, think that I should use the FRP system.	
<i>Top management support</i> Top management support is defined as the perceptions and actions of top officials on the usefulness of FRP in creating values for the organization (Intan Salwani et al., 2009).	TMS1 Top management in my organization believes that employees using the FRP system to provide customers with services is important and beneficial to the organization.	Wang & Lai (2014)
	TMS2 Top management in my organization strongly encourages employees to use the FRP system to provide customers with services.	
	TMS3 Top management in my organization is committed to supporting my efforts in using the FRP system to provide customers with services.	
	TMS4 Top management in my organization is willing to take risks involved in using the FRP system to provide customers with services.	

continued on following page

Table 1. Continued

Constructs	Measurement Items	Adapted from
Inertia Inertia refers to one's attachment to and persistence in using the current system even if there are new and better alternatives (Loh et al., 2021).	IN1 I will continue using the existing payment system to serve customers because it would be stressful to change to the FRP system.	Loh et al. (2021)
	IN2 I will continue using the existing payment system to serve customers because I am comfortable doing so.	
	IN3 I will continue using the existing payment system to serve customers because it is what I have always done.	
	IN4 I will continue using the existing payment system to serve customers because I have done it so regularly in the past.	
Perceived Value Perceived value refers to merchants' evaluation of a product or service based on their perception of the trade-off of sacrifices and benefits (Yang, 2020).	PV1 Compared to the effort I would need to put in, the change to using the FRP system for providing customers with services is beneficial to us.	Sohn & Kwon (2020); Tang et al. (2020)
	PV2 Balancing profits and losses, I think the change to using the FRP system for providing customers with services is worthwhile.	
	PV3 Considering the loss that I incur, the change to using the FRP system for providing customers with services is of good value.	
	PV4 Considering the hassle that I have experienced, the change to using the FRP system for providing customers is beneficial to me.	
	PV5 Overall, the change to using the FRP system delivers good value.	
Behavior intention Behavior intention is the subjective probability of an individual's engagement in a certain behavior (Fishbein & Ajzen, 1975).	BI1 I intend to use the FRP system to provide customers with services in the future.	Iranmanesh et al. (2023); Jum'a et al. (2022)
	BI2 I predict I will use the FRP system to provide customers with services in the future.	
	BI3 I plan to use the FRP system to provide customers with services in the future.	
	BI4 I would highly recommend the FRP system for my customers and my peers to adopt.	
Perceived customer value addition Perceived customer value addition is defined as the level of satisfaction of a service's customer, which is based on the customer's emotional, physical, and social response and perceived rating given to the service or retailer (Singh & Sinha, 2020).	PCVA1 The FPR system helps to improve our customer relationship based on the information collected.	Singh & Sinha (2020)
	PCVA2 The FPR system allows us to offer several benefits to our consumers (rewards/cash back/discounts, etc.).	
	PCVA3 The FPR system helps solve our customers' queries.	
	PCVA4 The FPR system makes the shopping experience pleasant for our customers.	
	PCVA5 The FPR system adds value to our consumers.	

Note. SWB: switching benefits; JIS: job insecurity; CO: colleague opinion; TMS: top management support; IN: Inertia; PV: perceived value; BI: Behavior intention; PCVA: perceived customer value addition.

into two parts. The first part collects basic information about the respondents, including gender, age, highest education level, job title, years of experience, industry type, and the type of service provided to consumers using FRP. The second part addresses switching benefits, job insecurity, colleagues'

opinions, top management support, inertia, perceived value, behavioral intention, and perceived customer value addition.

Sample and Data Collection

The subjects were employees of offline organizations with experience using FRP systems to serve customers. A pilot test involving five respondents who had experience with FRP systems offline was conducted to review and evaluate the logical consistency, semantics, length, sequence of items, and questionnaire format. Subsequently, based on the respondents' feedback, the questionnaire was revised to enhance comprehensibility and ensure content validity. Before the main investigation, data was collected from 163 respondents for the pretest, of which 57 (35.0%) were men and 106 (65.0%) were women. The factor loadings of all items in the pretest exceeded 0.5 (Wixom & Watson, 2001), indicating measurement accuracy. Concerning reliability evaluation, the composite reliabilities ranged from 0.819 to 0.929 (Fornell & Larcker, 1981), and Cronbach's alpha values ranged from 0.704 to 0.898. The average variance extracted (AVE) values, assessing convergent validity, all exceeded 0.5. The AVEs of constructs ranged from 0.503 to 0.766, indicating acceptable convergent validity (Fornell & Larcker, 1981). These results from the pretest confirmed the reliability and validity of this investigation.

Following the random sampling technique, data were collected using a self-administered structured questionnaire. After validating the pretest, the formal survey was conducted online, which offers several advantages over traditional paper-based surveys, including fast response time, a wide sample range, and cost efficiency (Bhattacharjee 2001a, 2001b). The questionnaire was distributed via Sojump (www.wjx.com), the earliest and largest online survey platform in China. The respondents were employees who have used FRP systems in offline settings to serve customers. The survey was conducted from December 2022 to February 2023. A total of 457 responses were collected. After screening incomplete and invalid responses (e.g., self-contradictory answers), 369 valid samples were retained for data analysis. The collected data were examined using inferential statistics by SEM, which helps to understand the population data well by analyzing the samples obtained from it. The sample covered various regions of China and included respondents from industries such as transportation, warehousing and logistics, postal services, wholesale and retail, accommodation and catering, financial services, leasing and business services, residential services, education, health, social security and welfare, and home building materials. This study takes into account the richness and diversity of the sample to interpret the FRP phenomenon.

RESULTS

Descriptive Statistics

The study sample consisted of 369 participants, with 167 (45.3%) identified as males and 202 (54.7%) as females. The majority of respondents fell within the age range of 20 to 39 years, accounting for 94.3% of the total sample. Regarding preferred customer payment methods, 55.0% of respondents indicated QR code payment as the most popular, followed by FRP systems. This finding aligns with the notion that FRP is a relatively new payment method in its initial commercial phase. Notably, a significant portion of organizations (67.0%) expressed favorable attitudes toward employees' utilization of FRP systems, either by encouraging preferential use or allowing voluntary adoption, while only a negligible proportion reported feeling coerced to use FRP systems. These results underscore the autonomy afforded to employees within organizations when utilizing FRP systems to serve customers.

Table 2 shows the statistics of the respondent's characteristics.

Table 2. Descriptive statistics of respondents' characteristics

Characteristics	Number	Percentage % (N=369)
<i>Gender</i>		
Female	202	54.7
Male	167	45.3
<i>Age</i>		
20–29	151	40.9
30–39	197	53.4
40–49	15	4.1
50–59	5	1.3
>60	1	0.3
<i>The highest education level</i>		
High school or below	9	2.4
Junior college	35	9.5
Bachelor's degree	298	80.8
Master's degree	26	7.0
Doctorate degree	1	0.3
<i>Currently, among the mobile payment options offered by your organization, which one do most customers choose?</i>		
FRP	158	42.8
QR code payment	203	55.0
Near-field communication payment	5	1.4
Point of sale payment	3	0.8
<i>What is the form of FRP service provided by your organization?</i>		
Only self-service facial recognition equipment service is provided (customers use self-service facial recognition machines for payment), with staff providing assisted instruction.	84	22.8
Only provide self-service facial recognition equipment service (customers use self-service facial recognition machine to pay), without staff providing assisted instruction.	40	10.8
Only manual cash register equipment services are provided (cashiers operate the facial recognition system to guide customers to pay).	32	8.7
At the same time, manual cash register service and self-service facial recognition equipment service (with staff providing assisted instruction) are provided.	203	55.0
At the same time, manual cashier service and self-service facial recognition equipment service are provided (without staff providing assisted instruction)	10	2.7
<i>What is your organization's attitude toward using the FRP system to provide customer payment services?</i>		
Mandatory command use	3	0.8
Encourage preferential use	247	67.0
Advocate voluntary use by employees	105	28.4
Noncommittal attitude	14	3.8

Nonresponse Bias and Common Method Bias

This study verified the absence of nonresponse bias following the method suggested by Armstrong and Overton (1977), which divided the data based on different response times. A total of 203 respondents who completed the early-stage survey were considered early respondents, whereas 166 respondents completed the late-stage survey. The chi-square test was employed to compare the sex and age distributions between the early and late respondent groups. The results indicate no significant differences in sex or age between the early and late respondents ($p > 0.05$). Therefore, the possibility of nonresponse bias can be confidently excluded.

In the survey, the authors anonymized the names of constructs and randomly assigned question items to mitigate common method bias (Podsakoff et al., 2003). Additionally, a common method bias test was conducted using various approaches. Harman's one-factor test was employed to identify any potential common method bias (Podsakoff & Organ, 1986). The results of Harman's test indicate that no single factor emerges alongside the primary factor, as required, with no single factor accounting for more than 50.0% of the variance (Mattila & Enz, 2002). The combined eight factors account for 58.5% of the total variance, with the largest factor explaining 31.7% (the explained variances ranged from 3.2% to 31.7%). Moreover, no general factor accounts for more than 50.0% of the variance. Additionally, common method bias was assessed using the marker variable technique (Chin et al., 2012). The data analysis results demonstrate that the marker variables exert no significant effects on perceived value and intention to use. The findings indicate that common method bias is not a concern in this study.

Measurement Model

The measurement model was employed to evaluate item reliability, internal consistency, convergent validity, and discriminant validity. All items demonstrated acceptable reliability, with factor loadings exceeding 0.6 (Wixom et al., 2001). Composite reliability and Cronbach's alpha were utilized to assess internal consistency reliability. Composite reliabilities ranged from 0.816 to 0.938, and Cronbach's alpha values ranged from 0.701 to 0.912, all surpassing the recommended threshold of 0.7 (Bagozzi & Yi, 1988). Convergent validity was evaluated using the AVE for each construct. The AVE values for all constructs exceeded 0.5, ranging from 0.523 to 0.791, indicating satisfactory convergent validity (Chin et al., 2003), as shown in Table 3.

Discriminant validity signifies the differentiation between constructs. To validate the discriminant validity of the constructs, both the Fornell–Larcker criterion and the heterotrait–monotrait ratio (HTMT) of correlation were evaluated. As depicted in Table 4, the square root of each construct's AVE exceeded its correlation with other constructs (Fornell & Larcker, 1981), and all HTMT of the correlations were below 0.9, the established threshold value (Henseler et al., 2015). With these two criteria satisfied, the discriminant validity of the model was confirmed.

Structural Model

Before examining the structural relationships, collinearity was assessed to ensure the elimination of bias. Variance inflation factor (VIF) values below the recommended threshold of 5.0 were considered acceptable (Neter et al., 1990). In this study, the outer VIF values ranged from 1.240 to 4.332, and the inner VIF values were all below 3.0. Therefore, these results indicated that collinearity was not a significant issue. To test the hypotheses, the bootstrap resampling technique (10,000 resamples) was employed to determine the significance of the structural model paths. It was found that switching benefits significantly and positively affected perceived value ($\beta = 0.276$, $t = 5.664$, $p < 0.001$) and behavioral intention ($\beta = 0.200$, $t = 3.529$, $p < 0.001$). Conversely, job insecurity significantly and negatively influenced perceived value ($\beta = -0.145$, $t = 3.262$, $p < 0.01$) and behavioral intention ($\beta = -0.128$, $t = 2.746$, $p < 0.01$). Colleague opinion significantly and positively impacted behavioral intention ($\beta = 0.169$, $t = 2.876$, $p < 0.01$) but not perceived value ($\beta = 0.098$, $t = 1.528$, $p > 0.05$). Moreover, top management support significantly and positively influenced perceived value ($\beta = 0.378$,

Table 3. Reliability and validity of the measurement model

Constructs	Items	Mean	SD	FL	AVE	CR
Switching benefits	SWB1	6.019	0.844	0.769	0.558	0.834
	SWB2	6.119	0.894	0.689		
	SWB3	5.379	1.086	0.725		
	SWB4	5.734	0.946	0.800		
Job insecurity	JIS1	2.648	1.340	0.827	0.695	0.932
	JIS2	2.509	1.334	0.865		
	JIS3	2.786	1.180	0.770		
	JIS4	2.810	1.265	0.866		
	JIS5	2.474	1.298	0.847		
	JIS6	2.542	1.335	0.825		
Colleague opinion	CO1	5.981	0.891	0.718	0.532	0.819
	CO2	5.954	0.875	0.705		
	CO3	5.550	1.071	0.763		
	CO4	5.615	1.084	0.729		
Top management support	TMS1	5.881	0.929	0.779	0.527	0.816
	TMS2	5.935	0.890	0.714		
	TMS3	5.967	0.962	0.746		
	TMS4	5.732	0.935	0.661		
Inertia	IN1	2.837	1.498	0.813	0.791	0.938
	IN2	3.553	1.621	0.906		
	IN3	3.558	1.696	0.926		
	IN4	3.412	1.665	0.909		
Perceived value	PV1	5.553	0.998	0.751	0.588	0.877
	PV2	5.870	0.948	0.757		
	PV3	5.696	1.028	0.751		
	PV4	5.699	1.046	0.784		
	PV5	5.911	0.988	0.791		
Behavior intention	BI1	6.062	0.812	0.811	0.584	0.849
	BI2	6.125	0.824	0.729		
	BI3	5.967	0.842	0.768		
	BI4	5.976	1.016	0.746		
Perceived customer value addition	PCVA1	5.355	1.007	0.747	0.523	0.846
	PCVA2	5.808	0.901	0.656		
	PCVA3	5.485	1.131	0.729		
	PCVA4	5.862	0.957	0.737		
	PCVA5	5.626	1.110	0.742		

Note. SWB: switching benefits; JIS: job insecurity; CO: colleague opinion; TMS: top management support; IN: Inertia; PV: perceived value; BI: Behavior intention; PCVA: perceived customer value addition; SD: standard deviation; FL: factor loading; CR: composite reliability.

Table 4. Discriminant analysis

Construct	SWB	JIS	CO	TMS	IN	PV	BI	PCVA
SWB	0.747							
JIS	−0.306 (0.370)	0.834						
CO	0.631 (0.876)	−0.298 (0.370)	0.729					
TMS	0.520 (0.721)	−0.313 (0.389)	0.620 (0.876)	0.726				
IN	−0.239 (0.287)	0.439 (0.480)	−0.209 (0.259)	−0.318 (0.396)	0.890			
PV	0.578 (0.742)	−0.377 (0.426)	0.549 (0.716)	0.627 (0.814)	−0.251 (0.288)	0.767		
BI	0.603 (0.799)	−0.434 (0.518)	0.594 (0.809)	0.597 (0.810)	−0.359 (0.428)	0.645 (0.810)	0.764	
PCVA	0.627 (0.832)	−0.338 (0.392)	0.622 (0.830)	0.634 (0.852)	−0.240 (0.277)	0.698 (0.872)	0.578 (0.743)	0.723

Note. SWB: switching benefits; JIS: job insecurity; CO: colleague opinion; TMS: top management support; IN: Inertia; PV: perceived value; BI: Behavior intention; PCVA: perceived customer value addition; The values on the diagonal (in bold and italic) are the square root of AVE for each construct, and the value in parentheses is HTMT ratio.

$t = 6.735$, $p < 0.001$) and behavioral intention ($\beta = 0.145$, $t = 2.172$, $p < 0.05$), while inertia significantly and negatively affected behavioral intention ($\beta = -0.106$, $t = 2.750$, $p < 0.01$). Additionally, perceived value was found to have a positive effect on behavioral intention ($\beta = 0.272$, $t = 3.229$, $p < 0.01$). All hypotheses were supported by the data except for H5. The use of the R square value to evaluate the structural model aids in interpreting the coefficient of determination (Hair et al., 2017). The model explains 50.4% of the variance in perceived value and 57.6% of the variance in behavioral intention.

The results of the structural model are depicted in Figure 2.

Moderating Effect Analysis

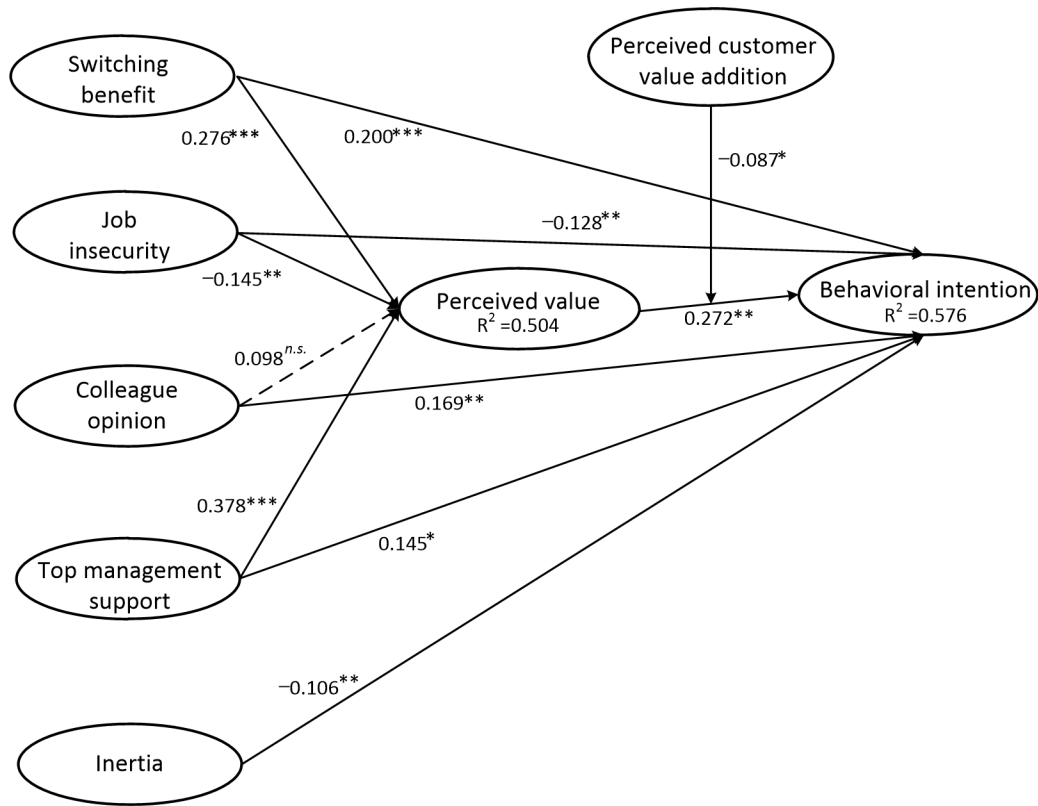
The moderating effect of perceived customer value addition was examined at the mean, with one standard deviation (SD) above the mean and one SD below the mean. The simple effects were verified, and H11 was rejected ($\beta = -0.087$, $t = 2.032$, $p < 0.05$). The simple slope diagram of perceived customer value addition indicates that the relationship between perceived value and behavioral intention increases more gradually among employee users with higher perceived customer value addition (perceived customer value addition at +1 SD) compared to those with lower perceived customer value addition (perceived customer value addition at −1 SD). This finding suggests that perceived customer value addition significantly and negatively moderates the effect of perceived value on behavioral intention.

Figure 3 presents the simple slope analysis of perceived customer value addition moderating the relationship between perceived value and behavioral intention.

Post-Hoc Rigorous Test in Endogeneity

The causes of endogeneity can be categorized into four main types: omitted variable, simultaneity, measurement error, and selection (Hill et al., 2020; Wooldridge, 2010). When addressing endogeneity issues, the instrumental variables approach is commonly employed and deemed suitable for mitigating endogeneity arising from various causes (Bastardo et al., 2023; Hult et al., 2018). Different estimation techniques are utilized based on specific conditions when applying the instrumental variables method

Figure 2. Results of the structural model



Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant

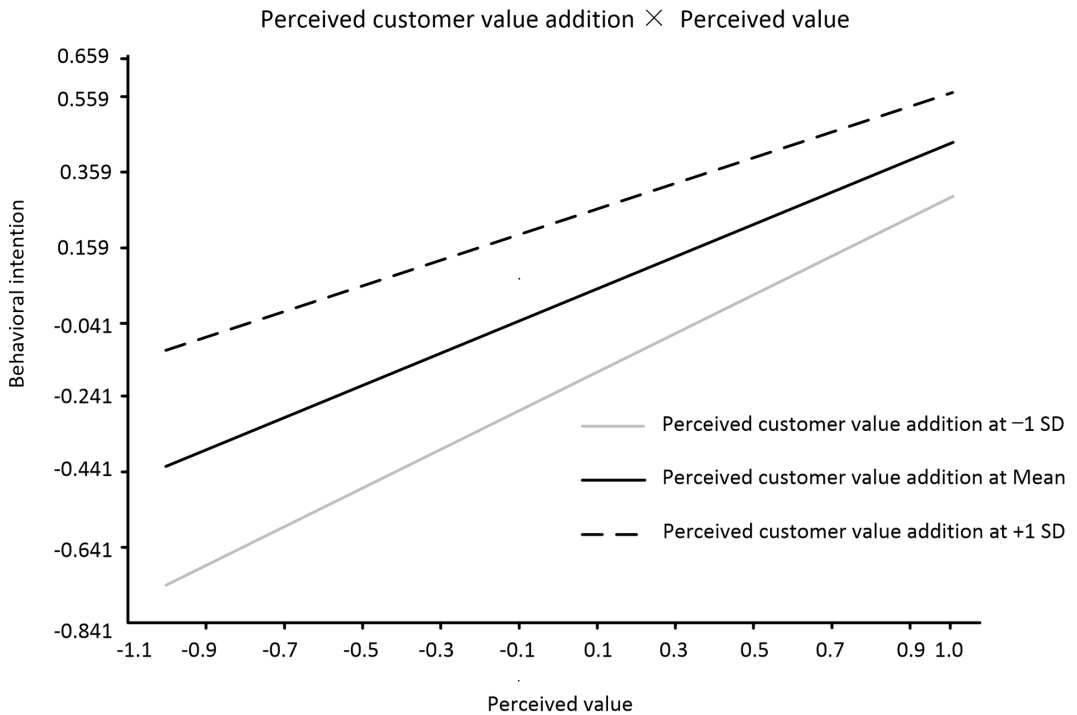
to address endogeneity concerns. For instance, studies have employed techniques such as two-stage least squares (Wei et al., 2023), generalized method of moments (Bojd et al., 2022), two-stage residual inclusion (Haislip et al., 2021; Wang & Ngai, 2022), and two-stage method of moments (Zhou et al., 2022). Among these, the most used approach to tackle endogeneity issues is the two-stage least squares regression (Bastardoz et al., 2023). To ensure this study's rigor, potential endogeneity concerns stemming from certain variables and selection biases were anticipated. Variables such as habit and personal innovativeness are notable candidates for causing endogeneity issues due to their relationships with other variables in the research model. Following the procedures outlined for addressing endogeneity in PLS-SEM (Hult et al., 2018), researchers should consider employing Gaussian copula, control variables, and instrumental variables.

Firstly, the Gaussian copula approach was utilized to identify potential endogeneity issues, revealing significant results in two relationships. Specifically, the relationship between job insecurity and behavioral intention ($\beta = 0.415$, $t = 2.039$, $p < 0.05$) and the relationship between colleague opinion and perceived value ($\beta = -0.229$, $t = 2.073$, $p < 0.05$) were found to be significant.

Secondly, control variables were incorporated to ascertain their influence. Despite the inclusion of sex, age, education, and industry type as control variables (Becker et al., 2016; Shiau et al., 2024), the Gaussian copula results for both relationships mentioned above remained significant.

Thirdly, habit and personal innovativeness were considered instrumental variables. Previous studies suggest that habit may influence job insecurity and intention, while personal innovativeness may positively affect subjective norms intention (De Kervenoael et al., 2021; Li et al., 2021). To

Figure 3. Simple slope analysis of the moderating effect



be deemed valid, instrumental variables should be correlated with the endogenous variable (job insecurity; colleague opinion) and unrelated to the dependent variables (behavioral intention; perceived value) error term. Consistent with recommendations to address endogeneity (Hult et al., 2018), the weak identification test's F-value was employed to assess the validity and strength of instrumental variables. Both habit and personal innovativeness were potential instrumental variables. The Hausman test (Hausman, 1978) was conducted to compare the coefficients of the original model and the instrumental variables model. When job insecurity was considered endogenous, the Hausman test yielded non-significant results (Hausman test $p > 0.05$, R-squared = 0.1783, Wald test $p < 0.001$), suggesting that habit is not a suitable instrumental variable. However, despite identifying a potential endogeneity issue based on the Gaussian copula result, the authors were unable to fully account for it using available control variables and instrumental variables.

Furthermore, personal innovativeness emerged as a potential instrumental variable. When colleague opinion was considered endogenous, the Hausman test yielded significant results (Hausman test $p < 0.001$, R-squared = 0.1924, Wald test $p < 0.001$), indicating that personal innovativeness is a weak instrumental variable. After including control variables and adhering to the requirements of instrumental variables, which stipulate that the direct effect on the dependent variable should be insignificant while the indirect effect is significant, it was found that personal innovativeness insignificantly influenced perceived value both directly ($\beta = 0.051$, $t = 0.876$, $p > 0.05$) and indirectly ($\beta = 0.047$, $t = 1.335$, $p > 0.05$). Consequently, personal innovativeness may not be a suitable instrumental variable for the relationship between colleague opinion and perceived value. Given the imperatives of reliability and validity, the authors strongly advocate for researchers to conduct rigorous post-hoc testing for endogeneity in future studies.

DISCUSSION

This study investigates how critical facilitating or inhibiting factors, acting as antecedents, influence employees' intention to use FRP systems or affect perceived value by integrating the VAM theoretical framework and SQB theory. Additionally, the moderating effect of perceived customer value addition on the relationship between value and intention is examined. The findings reveal that switching benefits positively influence perceived value and behavioral intention, thus supporting H1 and H2. Consistent with prior research, switching benefits facilitate adoption by offering direct or indirect advantages to users once the transition is established (Alzahrani et al., 2021; Chang, 2020). Employees view digital transformation as having greater value attached to their work tasks. In the context of FRP systems, adopting this new payment method can alleviate employees' workload, reduce the time spent serving customers, and enhance work efficiency. Consequently, employees perceive higher value from the new payment method compared to other alternatives, thus enhancing their intention to adopt it. Moreover, transitioning to a new payment system may lead to employees assuming new roles, further elevating their perceived value and intention to use. Furthermore, job insecurity negatively impacts perceived value and behavioral intention, thus supporting H3 and H4.

The introduction of new technology is a double-edged sword for employees. According to Lazarus and Folkman (1984), potential job loss is predicted to diminish positive outcomes, leading to emotional distress or efforts to safeguard job security. Consequently, job insecurity diminishes employees' evaluation of value and behavioral intention. This finding aligns with previous studies indicating that job insecurity may provoke resistance to change due to negative job attitudes (Feng et al., 2021; Shin et al., 2012). Additionally, its correlation with negative emotions such as stress, anxiety, and reduced well-being further explains the negative relationship between job insecurity and attitudes and behaviors (Aslam et al., 2016; Charkhabi, 2018).

This study's results indicate that colleague opinion does not significantly influence perceived value, rejecting H5, but positively impacts behavioral intention, supporting H6. The rejection of H5 may be attributed to the social influence factors' emphasis on conformity and imitation in the early stages of IS adoption (Venkatesh & Davis, 2000), which may not prominently impact benefit-sacrifice valence. However, colleague opinion directly affects employees' behavioral intention, contrasting with Kim and Kankanhalli's (2009) findings, where colleague opinion indirectly influences intention. This discrepancy may stem from the different research contexts. Unlike the comprehensive and complex systems studied by Kim and Kankanhalli (2009), the FRP system is relatively easy to learn and imitate, thus enabling colleague opinion to influence individual decision-making directly. This study's findings demonstrate that top management support positively influences perceived value and behavioral intention, supporting H7 and H8. These results are consistent with prior research, particularly in AI and robotics contexts, where top management support reduces interdepartmental conflicts and fosters organizational consensus (Pizam et al., 2022).

Top management plays a critical role in influencing and convincing members of the organization concerning the acceptance of a new technology (Kumar et al., 2017). Top management support strongly encourages FRP system usage by emphasizing its benefits and dispelling concerns. Consequently, with support from top management, employees perceive higher value and are more motivated to use FRP systems. Furthermore, top management support signifies a willingness to assume the risks associated with FRP, further bolstering employee usage intention.

Inertia negatively influences behavioral intention, supporting H9. This finding underscores the significant role of employee inertia in hindering the formation of behavioral intention toward FRP systems. Consistent with previous studies on various systems, including human resources management apps (Wang et al., 2018), bloggers' switching behavior toward microblogging (Cao et al., 2020), and new systems (Polites & Karahanna, 2012), employees are inclined to persist with current payment methods due to uncertainty about their comfort and familiarity with FRP systems. Moreover, perceived value positively impacts behavioral intention, supporting H10. This result indicates that employees

perceiving higher value from FRP systems are more willing to use and accept this new payment tool, consistent with previous literature on various technological innovations (Lau et al., 2019; Jun et al., 2018; Sandhu et al., 2023).

Interestingly, this study's unexpected results reveal that perceived customer value addition negatively moderates the relationship between perceived value and behavioral intention, rejecting H11. This suggests that higher perceived customer value addition does not significantly reinforce the alignment between employees' perceived value of FRP systems and their intention to use them. FRP is an advanced form of mobile payment that transforms how employees interact with consumers. This new interaction between consumers and technology offers a distinct experience. FRP systems, characterized by high intelligence and automated interaction, partially replace employees' job functions. Consequently, although employees recognize the significant value of FRP systems, they may perceive their roles as partially replaced when they observe that customers prefer interacting with FRP systems. Thus, the interaction between the value of FRP and perceived customer value addition may diminish employees' intention to use FRP systems. This finding is partially consistent with Hilton et al.'s (2013) study on organizations adopting self-service technology, which can blur the lines between customers and employees and introduce certain risks.

Theoretical and Practical Implications

This study has numerous theoretical and practical implications. Regarding theoretical implications, firstly, it integrates the SQB theory and VAM theoretical framework to discern the critical factors that either facilitate or inhibit employees' perceived value and behavioral intention within offline organizations. Employee intentions can be swayed by a bias or preference to maintain the current situation. Therefore, amalgamating technology acceptance with the SQB perspective sheds new light on implementing new IS. This framework incorporates rational decision-making, psychological commitment, and cognitive misperception—three explanatory dimensions of the SQB theory—various theories such as switching benefits, job insecurity, colleague opinion, top management support, and inertia are explored as significant influencing factors. Furthermore, this study integrates the theoretical framework into the offline service context of FRP. Secondly, this study delves into perceived customer value addition, which has garnered limited attention as a moderating variable. As the transition occurs gradually from employee-customer interaction to machine-customer interaction, employees' attitudinal evaluation and intention toward new technology or systems will be influenced by alterations in their perception of customer value. Thirdly, this study applies a rigorous endogeneity test to identify possible endogenous variables to the research model.

This study has numerous theoretical and practical implications. Theoretically, integrating SQB theory with the VAM framework allows for a more comprehensive view of the changes related to new IS within the context of technology acceptance. SQB reflects a general psychological phenomenon where the status quo choice acts as a psychological anchor. Consequently, a decision-making model incorporating status quo effects accurately describes economic behavior across various contexts (Samuelson & Zeckhauser, 1988). The integrated theoretical model thoroughly identifies the key psychological and social factors influencing employees' perceived value and behavioral intention through rational decision-making, psychological commitment, and cognitive misperception. Switching benefits and job insecurity reflect employees' most critical sacrifices and benefits. Colleague opinions and top management support represent the psychological commitment shaped by social norms and controls in the workplace. Inertia illustrates a common cognitive misperception among individuals. Furthermore, this study integrates the theoretical framework into the offline service context of FRP. It also explores perceived customer value addition, a moderating variable that has received limited attention. As the transition shifts from employee-customer to machine-customer interaction, employees' attitudinal evaluations and intentions toward new technologies will be influenced by changes in their perception of customer value. Perceived customer value addition reflects employees' subjective

judgments about customers' perceptions of FRP. Additionally, the post-hoc endogeneity test identifies potential endogenous variables in the research model.

In practice, the findings of this study offer several implications for management. Firstly, management should be cognizant of and mitigate the negative effects on employees' intention to use FRP. According to the findings, job insecurity negatively affects employees' intentions. Therefore, management should transparently communicate the purpose of introducing the FRP system to mitigate or reduce job insecurity among employees. If the implementation involves adjustments to employee positions, management should provide comprehensive explanations regarding the functions of these new positions and the future career trajectory of employees. Subsequently, after addressing employees' concerns and job insecurity, practical operational training, guidance, and resources should be extended to them. By lowering the operational threshold and nurturing employees' usage habits regarding FRP, they can overcome their inherent habits and inertia with external support. Secondly, according to the finding that top management support positively affects employees' perceived value and behavioral intention, management must effectively communicate the benefits of utilizing FRP systems to employees. Top management should unequivocally convey the organization's support for change and advocate for the adoption of the new technology system, thereby encouraging employees to embrace it. Such support fosters employees' intrinsic commitment and positively influences their value judgment and willingness to adopt the new technology. Moreover, top management needs to remain vigilant about the shifts in the interaction dynamics between customers and employees in the digital transformation. The perception of such changes will impact the relationship between employees' perceived value and their intention to use FRP.

Limitations and Future Research

Despite the previously mentioned implications, the current study has several limitations and future studies. First, in this study, the impact of differences in the type of industry the respondents work in remains to be studied. Future research should further explore the differences and commonalities of FRP systems adopted by employees in different industries. A comparative study on how acceptance of FRP varies among employees across different industries could offer valuable insights. Second, considering perceived customer value addition as a moderating variable should be applied in other service industries to capture employees' complex psychology. Third, the instrumental variables chosen at present did not achieve the expectation. Future research may consider other instrumental variables and also do a post-hoc rigorous endogeneity test. Fourth, this study focuses primarily on generalization and lacks qualitative analysis. Generalized research can explain social phenomena universally and completely. In the future, it is suggested that scholars further supplement the existing conclusions by conducting qualitative studies to gain deeper and richer insights. Fifth, as this study's samples are exclusively from China, there are limitations in generalizing the conclusions of this study to a global context. Future research could involve similar or more intensive studies in different countries and regions. Sixth, this study presents limited technical details. For specific technical information, please refer to the books by Hair et al. (2017) and Hair et al. (2009). Additionally, investigating samples from other FRP application areas could provide an in-depth understanding and further enrich the research findings on FRP.

CONCLUSION

With the organization's digital reform evolving and reshaping the roles of employees in customer interactions, it is imperative to delve into the crucial influencing factors affecting employees' behavioral intention toward FRP systems. A total of 369 valid samples were collected and analyzed using PLS-SEM. The research model elucidates 50.4% of the variance for perceived value and 57.6% for behavioral intention. Results indicate that switching benefits and top management support positively influence perceived value and behavioral intention. Colleague opinion exhibits a significantly negative

effect solely on behavioral intention, not perceived value. Job insecurity negatively impacts both perceived value and behavioral intention. Inertia only detrimentally affects behavioral intention. Perceived value positively influences behavioral intention. Additionally, perceived customer value addition significantly moderates the relationship between perceived value and behavioral intention, exerting a negative effect.

This study develops a theoretical model for employee technology adoption by integrating the VAM and SQB theoretical perspectives to elucidate employees' decision-making strategies. This model offers a comprehensive estimation of the perceived value of FRP systems, while the SQB theoretical perspective identifies critical factors that either facilitate or inhibit adoption as antecedents. Moreover, the revelation of the moderating effect of perceived customer value addition sheds light on deeper employee concerns. Furthermore, the endogeneity tests enhance the rigor of this study. Traditional organizations are transitioning digitally to “do more with less” (Hilton et al., 2013), incorporating FRP systems to reduce labor costs while enhancing service quality and experience. Expanding on these findings, there is a compelling prospect of utilizing the integrated framework of VAM and SQB theory not solely to investigate employees' inclination toward adopting FRP systems but also to address diverse digital transformation scenarios. This endeavor holds promise for augmenting specialized knowledge in the field.

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COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

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